INCH-POUND

MIL-C-24643/15D
22 November 1994
SUPERSEDING
MIL-C-24643/15C
14 March 1994

MILITARY SPECIFICATIONS SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSDSGU
(INCLUDING VARIATION LSDSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualifications required.

Construction (watertight with circuit integrity)

Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

First - Copper conductor, uncoated, (see table I for size)
Second - Extruded Silicone rubber insulation (see table I for diameter).
Third - Glass braid. 80% minimum coverage.
Fourth - Braid covering (see table I for diameter). Standard identification code applied by method 1 or method 3.

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation. 1/

First - Copper conductor, uncoated, (see table I for size)
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
Third - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fourth - A binder.

1/ Size 23 through 100 may be glass tape or excluded silicone rubber insulation.

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TABLE I. Details.

Military part no. M24643/15 1/	Size 2/	Conductor size		Type's overall diameter (inches)				Diameter over outer insulation	Diameter over braid	Diameter over Insulation	Jacket thickness minimum	Conductor resistance	Insulation resistance	Cold bending mandrel diameter	Accelerated service loading
		Navy std	AWG	LSDSGU		LSDSGA		Tape nom.	maximum	minimum	average			maximum	
				min	max	min	max	(inch)	(inch)	(inch)	(inch)	(ohms)	(megohms)	(inch)	ampere
-01UN	3		16 Cl B		0.391		0.441		0.130	0.096	0.030	4.3	500		
-02UN	4		14 Cl B		.427		.477		.143	.112	.030	2.68	500		
-03UN	9		10 Cl B		.544		.594	----	.187	.154	.040	1.08	500	----	----
-04UN	14		9 Cl B		.670		.720		.262	.230	.040	.859	500		100
-05UN	23		7 Cl B		.781		.831	0.274	.310	.271	.050	.543	500		130
-06UN	50		3 Cl C		.911		.961	.334	.380	.335	.050	.21	200	12	215
-07UN	75		1 Cl C		1.074		1.124	.407	.440	.407	.050	.134	175	13	285
-08UN	100		0 Cl D		1.167		1.217	.453	.500	.453	.050	.106	160	15	335
-09UN	200		0000 Cl D		1.583		1.633	.634			.060	.053	125	20	525
-10UN	300		300 MCM Cl D		1.841		1.891	.748			.075	.0377	110	23	700
-11UN	400	400 (127)			2.069		2.119	.862			.075	.0273	100	25	835

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LSDSGU and LSDSGA

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Construction (watertight and circuit integrity) (continued)

Construction variant type LSDSGU

Sizes 3 to 100 inclusive, extruded silicone rubber insulation 1/

Fifth - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly
Sixth - Binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. (see table I for thickness).

Sizes 23 to 400 inclusive, silicone rubber-glass tape insulation 1/

Fifth - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant type LSDSGA

Fifth - Cross-linked polyolefin jacket. (see table I for thickness).
Sixth - Braided metal armor.

1/ Size 23 through 100 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

EXAMINATION AND TESTS

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum...	(see table I)
Voltage withstand - volts, root mean square, minimum	
Sizes 3 through 9	3000
Sizes 14 and larger.....	5000
Insulation resistance - conductor to conductor, megohms/1000 feet, minimum	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
* Abrasion resistance - sizes 3 through 100 (extruded silicone rubber) scrapes, minimum	250
* Crack resistance - sizes 3 through 100 (extruded silicone rubber)	No damage

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EXAMINATION AND TESTS (continued)

Group B:

	Requirements
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

Group C:

Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (conductor, method 1 only) - cycles, minimum (extruded silicon rubber only)	25
* Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Armor (type LSDSGA only) - conformance to material, construction and coverage	No failure

Group D:

Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ± 2°C)	No damage
Gas Flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Accelerated service - sizes 14 and larger (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSDSGU-3	2500
LSDSGU-4 through 23	2000
LSDSGU-50	1500
LSDSGU-75 through 100	1000
LSDSGU-200 through 400	800

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(Project No. 6145-2101-15)

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